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EDUCATION

Master of Arts <i>Policy Economics</i> Williams College	2016 – 2017 USA
Master of Science <i>Applied Economics</i> Universidad Andina Simón Bolívar	2015 – 2016 Bolivia
Bachelor's Degree <i>Economics</i> Universidad Mayor de San Andrés	2006 – 2010 Bolivia

WORK EXPERIENCE

Consultant REMOTE JOB	Open to consulting opportunities
• 2026 ESTIMATING THE CAUSAL IMPACT OF BROADBAND SPEED ON POVERTY, INCOME, AND LABOR INFORMALITY IN BRAZIL AND ECUADOR [difference-in-differences a la Callaway–Sant’Anna; geospatial methods]; project funded by the <i>United Nations Development Programme</i> (UNDP). • 2026 DESIGNING A FRAMEWORK TO RESTRUCTURE INTERGOVERNMENTAL TRANSFERS IN BOLIVIA [microsimulation of alternative transfer formulas across municipalities]; project funded by the <i>Inter-American Development Bank</i> (IDB). • 2026 CO-AUTHOR RESEARCHER WITH UPB-SDSN ON CLIMATE SHOCKS AND SUBNATIONAL PUBLIC FINANCES IN BOLIVIA [Local Projection; satellite-derived climate shocks]; Fiscal Tools to Manage Climate-Induced Disaster Risk project; funded by the <i>Inter-American Development Bank</i> (IDB). • 2026 CO-AUTHOR RESEARCHER WITH FUNDACIÓN MILENIO ON EFFECTIVE ECONOMIC FREEDOM AND ENTERPRISE GROWTH IN BOLIVIA [firm-level microeconometrics with the EMyPE survey]; project funded by the <i>Center for International Private Enterprise</i> (CIPE). • 2020 – 2021 ASSESSED THE ECONOMIC IMPACT OF TRANSPORT INFRASTRUCTURE IN BOLIVIA, ECUADOR, AND PARAGUAY [econometric and remote sensing methodologies]; Economic Integration in Latin America Project, funded by the <i>Development Bank of Latin America</i> (CAF). • 2018 EVALUATED LOCAL PUBLIC INVESTMENT EFFECTIVENESS AND ITS SPILLOVER EFFECTS ON DEVELOPMENT OUTCOMES [spatial econometric techniques]; Strengthening Economic Development Research Project, funded by the <i>Development Bank of Latin America</i> (CAF).	

Onsite Experience

Head of Economic Affairs CAINCO / BOLIVIAN CENTER OF ECONOMICS	Jan. 2026 – Present Bolivia
• Lead data analytics, Big Data initiatives, and economic research utilizing advanced econometrics, machine learning, and generative AI techniques. • Generate macroeconomic and sectoral projections encompassing the real, external, fiscal, and monetary-financial sectors.	
Lead Specialist in Artificial Intelligence for Economic Research MINISTRY OF ECONOMY AND PUBLIC FINANCE	Feb. 2024 – Nov. 2025 Bolivia
• Directed AI and data science initiatives to support evidence-based economic policymaking (e.g., predictive models and advanced analytics to enhance real-time macroeconomic monitoring and decision-making). • Proposed, secured funding for, and led the establishment of an AI Laboratory within the Ministry, in collaboration with the IDB.	

Lead Researcher MINISTRY OF ECONOMY AND PUBLIC FINANCE	May 2021 – Feb. 2024 Bolivia
- Led research projects on socioeconomic trends and policy evaluation, utilizing advanced econometrics, machine learning, and big data analytics. - Provided direct technical advice to senior leadership, including the Minister of Economy. - Published peer-reviewed research articles in the Ministry's economic journal.	
Economist, Economic Modelling UNIDAD DE ANÁLISIS DE POLÍTICAS SOCIALES Y ECONÓMICAS (UDAPE)	Aug. 2017 – Feb. 2021 Bolivia
- Led modelling and analysis of macroeconomic and social policies, designing advanced econometric tools to support policy formulation. - Coordinated multi-institutional research with international organizations (CAF, UNFPA).	
Researcher MINISTRY OF ECONOMY AND PUBLIC FINANCE	Jan. 2015 – Aug. 2016 Bolivia
Economist, Real Sector UNIDAD DE ANÁLISIS DE POLÍTICAS SOCIALES Y ECONÓMICAS (UDAPE)	Apr. 2013 – Dec. 2014 Bolivia
Economist, Fiscal Sector UNIDAD DE ANÁLISIS DE POLÍTICAS SOCIALES Y ECONÓMICAS (UDAPE)	Oct. 2012 – Dec. 2012 Bolivia
Technical Assistant, Fiscal Sector UNIDAD DE ANÁLISIS DE POLÍTICAS SOCIALES Y ECONÓMICAS (UDAPE)	Sept. 2011 – Sept. 2012 Bolivia

PEER-REVIEWED ARTICLES / BOOK CHAPTERS / WORKING PAPERS

- Bolivar, O., Canavire-Bacarreza, G. & Balthrop, A. (2026). [Hit the Road Juan: Welfare and Labor Market Effects of High-Quality Roads in Ecuador](#). *IZA Discussion Papers — IZA@LISER Network*.
- Bolivar, O., Yarygina, A. & Radics, A. (2026). [When Population Counts: Intergovernmental Transfers and Local Fiscal Performance in Bolivia](#). *IDB Working Paper in progress*.
- Bolivar, O. (2025). [High-frequency Inflation Forecasting: A Two-step Machine Learning Methodology](#). *Latin American Journal of Central Banking*, 7(1), 100172.
- Bolivar, O. & Mendez, C. (2025). [Predicting community-level poverty in Bolivia: Insights from satellite imagery, census data, and spatial modeling](#). *Nagoya University (Ed.), Revisiting the Poverty-Growth-Inequality Triangle in the Post-Pandemic Era (in progress)*. Japan: Routledge – Taylor & Francis Group.
- Bolivar, O. (2025). [Fire Vulnerability Mapping in the Chiquitano Biome: A Machine Learning Approach with MapBiomass Data](#). *Available at SSRN*.
- Bolivar, O. & Huanto, C. (2025). [Sentiment, Cryptocurrency and Inflation: A Transmission Channel in Bolivia](#). *Available at SSRN*.
- Bolivar, O. (2024). [GDP Nowcasting: A Machine Learning and Remote Sensing Data-based Approach for Bolivia](#). *Latin American Journal of Central Banking*, 5(3), 100126.
- Bolivar, O. & Huanto, C. (2024). [Tensiones Geopolíticas: Efectos Macroeconómicos en Bolivia, Chile y Perú](#). *Proceeding Paper - Call for Papers: 2024 Meeting of Economists of Bolivia*, Central Bank of Bolivia.
- Bolivar, O. & Huanto, C. (2023). [Geopolitical Risk Shocks: Macroeconomic Effects in Bolivia, Chile, and Peru](#). *Cuadernos de Investigación Económica Boliviana*, 6(2): 62-83.
- Bolivar, O., Huanto, C. & Teran, R. (2023). [Inflation Expectations: Dynamics and Effects for South America](#). *Cuadernos de Investigación Económica Boliviana*, 6(1): 3-24.
- Bolivar, O. (2023). [Evolución de la pobreza en las comunidades de Bolivia entre 2012 y 2022: Un enfoque de machine learning y teledetección](#). *MPRA Paper from University Library of Munich*.
- Bolivar, O. (2022). [Fiscal Multipliers: Empirical Evidence for a Cost-Effective Allocation of Public Investment](#). *Cuadernos de Investigación Económica Boliviana*, 5(2): 1-28.

- Bolivar, O. (2022). **Dinámica Inflacionaria, Crecimiento Económico y Empleo en los contextos de pandemia y pospandemia para América del Sur**. *Proceeding Paper - First Place in the Call for Papers: 2022 Meeting of Economists of Bolivia*, Central Bank of Bolivia.
- Bolivar, O. (2022). **Roads illuminate development: Using nightlight luminosity to assess the impact of transport infrastructure**. *Research Department Working Papers 1910*, CAF - Development Bank of Latin America.
- Bolivar, O. (2021). **Boyanza Tributaria: Relación con la sostenibilidad fiscal y estabilizadores automáticos**. *Cuadernos de Investigación Económica Boliviana*, 4(2): 53-95.
- Bolivar, O. (2020). **Evaluación de impacto de la infraestructura vial en el crecimiento económico: Una aproximación con base en información satelital de luminosidad para Bolivia**. CAF-UDAPE, Osmar Bolívar (Ed.), *Evaluación de impacto de la infraestructura vial en Bolivia* (pp. 11-70). La Paz, Bolivia.
- Bolivar, O. & Ugarte, D. (2020). **Evaluación de impacto de la infraestructura vial sobre las condiciones socioeconómicas de la población**. CAF-UDAPE, Osmar Bolívar (Ed.), *Evaluación de impacto de la infraestructura vial en Bolivia* (pp. 71-117). La Paz, Bolivia.
- Ugarte, D. & Bolivar, O. (2018). **Descentralización en Bolivia: Eficacia y Externalidades**. CAF-INESAD-ABCE & Beatriz Muriel (Ed.), *Investigaciones para el Desarrollo Económico de Bolivia* (pp. 113-168). La Paz, Bolivia.
- Bolivar, O. (2017). **Desdolarización en Bolivia: Interacción con la Política Monetaria y el Tipo de Cambio**. *Proceeding Paper - Call for Papers: 2017 Meeting of Economists of Bolivia*, Central Bank of Bolivia.
- Bolivar, O. (2016). **Bolivianization (De-dollarization) and Effectiveness of the Monetary Policy**. *Cuadernos de Investigación Económica Boliviana*, 1(2): 1-40.
- Bolivar, O. & D. Ugarte (2015). **Domestic Demand as Engine of the Economic Growth in Bolivia**. *Cuadernos de Investigación Económica Boliviana*, 1(1): 1-27.
- Ugarte, D. & Bolivar, O. (2015). **The Effect of Income Redistribution on Poverty Reduction of Bolivia**. *Cuadernos de Investigación Económica Boliviana*, 1(1): 1-25.
- Ugarte, D. & Bolivar, O. (2015). **La Relación Precio del Petróleo y Crecimiento Económico en Bolivia: El Rol de la Política Económica**. *Revista de Análisis del Banco Central de Bolivia*, 22(1): 9-40.

HONORS AND AWARDS

Winner & Funding, 2024 Research Proposal Call, Development Bank of Latin America (CAF). *For the project: "Digital Highways: The Causal Impact of Internet Speed on Poverty and Informality in Ecuador and Brazil"*

First Place, 2023 Central Bank Award, Center for Latin American Monetary Studies (CEMLA). *For the research: "GDP Nowcasting: A Machine Learning and Remote Sensing Data-Based Approach for Bolivia"*

First Place, 2022 Meeting of Economists of Bolivia, Central Bank of Bolivia. *For the research: "Dinámica Inflacionaria, Crecimiento Económico y Empleo en los Contextos de Pandemia y Pospandemia para América del Sur"*

Finalist & Funding, 2020 Research Proposal Call, Development Bank of Latin America (CAF). *For the project: "Roads Illuminate Development: Using Nightlight Luminosity to Assess the Impact of Transport Infrastructure"*

Winner & Funding, 2018 Research Proposal Call, Development Bank of Latin America (CAF). *For the project: "Descentralización en Bolivia: Eficacia y Externalidades"*

First Place, 2015 Meeting of Economists of Bolivia, Central Bank of Bolivia. *For the research: "La Relación Precio del Petróleo y Crecimiento Económico en Bolivia: El Rol de la Política Económica"*

TEACHING EXPERIENCE

Postgraduate Professor
Business School - Universidad Autónoma Gabriel René Moreno

2026 - Current
Bolivia

CERTIFICATIONS AND COURSES

Certificate Program in Applied Generative AI — Johns Hopkins University	2025
Deep Learning Specialization — DeepLearning.AI	2024
Machine Learning Specialization — DeepLearning.AI & Stanford University	2023
Multidisciplinary Program in Data, Programming, and AI — Buenos Aires University	2022
Training Course on Space Technology for Bolivia — Wuhan Research Institute of Posts and Telecommunications (China)	2022
R and GIS (Programming in R for Geographic Information Systems) — MappingGIS, Spain	2020
Remote Sensing in QGIS (Satellite Imagery Processing) — MappingGIS, Spain	2019
Dynamic Stochastic General Equilibrium Models — International Monetary Fund	2018

SKILLS

Languages: Spanish (Native), English
Programming/Software: Python, R, QGIS, STATA, MATLAB, E-views
Document Creation: Microsoft Office, LaTeX, Markdown